

NESTORA

AI Bathroom Studio

AI Prompt & Workflow Documentation

Comprehensive record of the generative-AI prompts, structured outputs, guardrails and deterministic workflows used in the prototype.

CORE PRINCIPLE | AI reasons. The spatial engine validates.

Submission context

Prepared as technical documentation for the KOHLER AI Lab Program. The document describes what Gemini is asked to do, what information it receives, how its output is validated, and which parts of Nestora deliberately remain deterministic.

AI model	Gemini 2.5 Flash
Backend	FastAPI / Python
Spatial layer	Deterministic geometry and validation
Visualization	SVG 2D + Three.js 3D
Primary AI roles	Vision analysis, grounded design reasoning, natural-language refinement

1. Role of AI in Nestora

Nestora does not ask a language model to freely generate bathroom coordinates. Instead, AI is placed around a deterministic planning core. This makes the AI useful for interpretation and explanation while keeping measurable placement decisions testable in code.

Where Gemini is used

Capability	AI responsibility	Planner responsibility
Room-image analysis	Interpret visible room features and provide cautious structured estimates.	Accept dimensions only when marked reliable; otherwise retain manual measurements.
Design intelligence	Explain computed layout facts and trade-offs in concise natural language.	Compute the facts, metrics, strategy and selected product data first.
Natural-language refinement	Translate a request such as 'more storage' into allowed planner preferences.	Validate values and regenerate spatial candidates under deterministic constraints.

Boundary of responsibility: Gemini can interpret intent and explain evidence. It is not treated as the authority for room fit, fixture collisions, entrance clearance, product existence or compliance.

2. System Workflow

Step	Layer	Operation
1	User input	Room dimensions, door position, budget, theme, design priorities and optional image.
2	AI interpretation	Optional image understanding or natural-language intent extraction.
3	Spatial engine	Generate candidate layouts and compute occupied footprints.
4	Validation	Check room bounds, entrance keep-clear zone and placement constraints.
5	Product / budget logic	Build a catalog bundle within the target budget.
6	AI design review	Provide Gemini only the computed facts required to explain the selected design.
7	Presentation	The validated state drives 2D, 3D, saved designs and PDF output.

3. Prompt A - Bathroom Image Analysis

Endpoint: POST /api/analyze-image Model: gemini-2.5-flash Response: JSON

Objective

Extract planning-relevant observations from an uploaded bathroom or room photograph without pretending that an ordinary image provides guaranteed real-world measurements.

Exact prompt

```
Analyze this bathroom/room image. Return JSON only with:
length_ft: estimated room length from 6 to 20;
width_ft: estimated room width from 5 to 16;
suggested_theme: one of modern_cozy, scandinavian, boho, classic, mediterranean,
minimalist, rustic, luxury_spa;
door_position: bottom_left or bottom_right;
confidence: number 0 to 1;
dimensions_reliable: boolean;
detected_features: array of short strings such as toilet, vanity, shower, tub, window;
notes: short string.
Only mark dimensions_reliable true when the image contains a credible scale reference.
Otherwise provide a rough estimate but explicitly mark it unreliable.
```

Inputs and expected output

Input	Uploaded image bytes.
Expected fields	length_ft, width_ft, suggested_theme, door_position, confidence, dimensions_reliable, detected_features, notes.
Downstream use	Theme/door suggestions may update the setup; dimensions are applied only when the response marks them reliable.
Failure behavior	If analysis is unavailable, the application leaves manual dimensions unchanged and reports the failure.

Vision safeguard: exact room dimensions cannot be guaranteed from a single photograph without a known scale reference. The prompt explicitly requires unreliable estimates to be labelled as such.

Illustrative example

User action: uploads a bathroom image showing a vanity, toilet and walk-in shower, but no trustworthy scale reference.

```
{
  "length_ft": 10.0,
  "width_ft": 8.0,
  "suggested_theme": "modern_cozy",
  "door_position": "bottom_left",
  "confidence": 0.72,
  "dimensions_reliable": false,
  "detected_features": ["toilet", "vanity", "shower"],
  "notes": "Room features detected; dimensions require manual confirmation."
}
```

The example above illustrates the schema and intended behavior; it is not a recorded model response from a specific submission run.

4. Prompt B - AI Design Intelligence

Endpoint: POST /api/designs/intelligence **Model:** gemini-2.5-flash **Response:** JSON

Objective

Turn already-computed design evidence into a concise review. The model explains the design; it does not create the measurements it discusses.

System instruction

You are Nestora's bathroom design reasoning layer.
Use ONLY the supplied computed facts and user brief.
Do not invent measurements, compliance claims, or products.
Return JSON with keys:
summary (2 sentences),
tradeoff (1 sentence),
next_action (1 sentence).

Grounding context supplied to Gemini

brief	User priorities such as storage, circulation, bath preference, accessibility and plumbing flexibility.
facts	Measurements and metrics computed by the application.
comparison	Evidence comparing generated layout candidates.
measured_tradeoff	A concrete trade-off derived from the selected layout.
layout / strategy	The selected spatial arrangement and strategy family.
products	Selected catalog product names, categories and prices.

Grounding rule: the prompt explicitly prohibits invented measurements, compliance claims and products. This keeps the narrative tied to application evidence.

Illustrative example

```
{
  "summary": "The selected layout keeps the main circulation path open while grouping the vanity-side fixtures efficiently. It also preserves the entrance keep-clear zone.",
  "tradeoff": "The storage-first arrangement uses more wall area than the open-balanced option.",
  "next_action": "Review the selected product bundle and confirm room measurements before installation."
}
```

This is a documentation example showing the required response shape, not a claim that these exact sentences were returned during evaluation.

5. Prompt C - Natural-Language Design Refinement

Endpoint: POST /api/designs/refine **Model:** gemini-2.5-flash **Response:** constrained JSON

Objective

Convert a short redesign request into planner preferences without allowing free-form model output to directly alter fixture coordinates.

Exact prompt

```
Convert the bathroom redesign request into JSON only.
Allowed keys:
storage_priority (low|medium|high),
circulation_priority (medium|high),
plumbing_flexibility (limited|flexible),
bath_preference (shower|tub|both),
accessibility (standard|step_free|enhanced).
Omit anything not requested.
Request: [user instruction]
```

Accepted values

storage_priority	low medium high
circulation_priority	medium high
plumbing_flexibility	limited flexible
bath_preference	shower tub both
accessibility	standard step_free enhanced

The returned object is checked against explicit allow-lists before it is accepted. The application also handles clear keyword patterns before Gemini, including storage requests, open-circulation requests, keep-plumbing requests, shower/tub preferences and accessibility language.

Important: the refinement response changes planner constraints. It does not move a fixture itself. Layout regeneration and validation remain deterministic.

Illustrative examples

User request	Expected constrained interpretation
"Give me more storage but keep the plumbing where it is."	{"storage_priority":"high","plumbing_flexibility":"limited"}
"Make it easier for wheelchair access."	{"accessibility":"enhanced","circulation_priority":"high"}
"I want a bathtub instead."	{"bath_preference":"tub"}

6. Deterministic Workflows - No LLM Prompt

The following functions are intentionally performed by application logic rather than a generative model. This is a core part of Nestora's technical design.

Workflow	Deterministic responsibility
Layout generation	Generate candidate fixture placements from room dimensions, fixture dimensions, door position and strategy.
Footprint normalization	Recompute occupied width/depth from catalog dimensions and rotation.
Boundary enforcement	Keep the occupied footprint inside the room.
Entrance protection	Preserve the planner's entrance keep-clear region.
Layout validation	Check placement state separately from AI commentary.
Spatial suggestions	Use entrance/swing zone, closest fixture gap and floor occupancy to generate measured observations.
Budget optimizer	Select mandatory core products, add affordable optional items, then upgrade while staying within budget.
2D / 3D synchronization	Drive both views from the same current layout state.
Saved design / PDF	Persist and export the validated design state.

Why this matters: a fluent AI explanation is not evidence that a physical layout fits. Nestora keeps spatial correctness in code so that it can be inspected, validated and tested.

7. Layout Strategy & AI Interaction

The planner can generate multiple strategy families such as **open_balanced**, **storage_first**, **spa_corner**, **symmetry**, **circulation_first** and **compact_comfort**. User design-intelligence preferences influence which families are emphasized, while the spatial engine remains responsible for actual placement.

Preference signal	Typical planner effect
High storage priority	Emphasize storage-first candidates.
High circulation priority	Emphasize circulation-first and open-balanced candidates.
Enhanced / step-free access	Prioritize circulation-oriented candidates.
Limited plumbing flexibility	Favor decisions that keep plumbing changes constrained.
Shower / tub / both	Influence the required bathing fixture category.

8. Guardrails, Validation & Failure Handling

Nestora uses several layers of control so that model output does not silently become spatial truth.

Control	Implementation intent
JSON-only prompting	Reduces ambiguous free-form output where application state is involved.
Explicit schemas / allowed values	Limits model responses to fields the planner understands.
Reliability flag for vision	Prevents rough image estimates from being treated as confirmed measurements.
Grounded design-intelligence prompt	Restricts commentary to supplied facts, comparison data and catalog products.
Deterministic validation	Checks geometry independently of AI-generated text.
Graceful fallback	Planner functionality can continue when Gemini is unavailable; image analysis reports failure instead of overwriting manual measurements.

9. API Key & Data Handling

Server-side key: Gemini requests are made from the backend using the GEMINI_API_KEY environment variable. The application does not intentionally place that secret in browser-side JavaScript.

Uploaded image: image data is supplied to the model for the requested room-analysis operation. If the AI call is unavailable or fails, the interface reports that result and keeps manually entered dimensions intact.

Prototype limitation: Nestora is a design-planning prototype. AI output and generated layouts should not be treated as professional construction, code-compliance or installation approval. Real room dimensions and installation requirements should be independently confirmed.

Submission principle: document the AI that actually exists. Nestora does not pad the system with fictitious prompts; the three model workflows above are the principal Gemini prompt paths in the prototype.

10. Prompt Inventory & Traceability

Capability	Endpoint	Model / method	Primary control
Bathroom image analysis	/api/analyze-image	Gemini 2.5 Flash	JSON schema + dimensions_reliable safeguard
Design intelligence	/api/designs/intelligence	Gemini 2.5 Flash	Use only supplied facts; no invented measurements/products
Natural-language refinement	/api/designs/refine	Gemini 2.5 Flash	Allowed keys + allowed-value validation
Spatial suggestions	/api/designs/suggestions	Deterministic code	Measured spatial observations; no LLM prompt
Layout validation	/api/layout/validate	Deterministic code	Geometry validation; no LLM prompt

11. End-to-End Example Workflow

Stage	Example
Brief	Two users; high storage; shower preferred; high circulation; limited plumbing movement.
Optional image analysis	AI detects visible features and suggests a theme/door position; uncertain dimensions remain unconfirmed.
Candidate generation	Spatial engine creates several strategy-based layouts.
Validation	Room bounds, entrance zone and fixture footprints are checked.
Bundle selection	Catalog products are selected within the chosen budget.
Design review	Computed facts and the selected bundle are sent to the grounded design-intelligence prompt.
Refinement	User says 'more storage'; constrained preferences are updated and candidates are regenerated.
Final output	The current validated layout drives the 2D plan, 3D room, saved design and PDF.

12. Submission Summary

Nestora's AI architecture is intentionally hybrid. Gemini contributes visual understanding, structured interpretation of natural language and evidence-grounded design commentary. Deterministic application logic retains responsibility for geometry, layout validation, budget logic and visualization state.

AI reasons. The spatial engine validates.

This document records the principal AI prompts and workflows present in the submitted prototype and distinguishes them from deterministic application behavior.